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Parity-Based Level-Set Approach to the Collatz Conjecture

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Abstract

The Collatz conjecture concerns the iteration of the map $f(n) = 3n+1$ for odd n and $f(n) = n/2$ for even n. In this paper, we study the level sets $l_x = \{n \in \mathbb{N} \mid L(n) = x\}$, where $L(n)$ denotes the Collatz length. Using the parity representation of Collatz trajectories, we partition each l_x according to the number of odd steps and analyze the corresponding means $\mu_{x,k}$. Under a natural scaling assumption, these means satisfy an approximate geometric progression, so that $\log \mu_{x,k}$ is approximately linear in k. Computations for $n \leq 100{,}000$ and $10 \leq x \leq 50$ show highly stable regression parameters and near-perfect linear fits.

Keywords: Collatz length; level sets; parity code; mean; regression; clustering

MSC: 11B25; 62E10; 62J05

1. Introduction

The Collatz conjecture, also known as the $3x+1$ problem, is one of the most well-known unsolved problems in mathematics. It concerns the iteration of the map $f : \mathbb{N} \to \mathbb{N}$ defined by

$$f(n) = \begin{cases} 3n+1, & \text{if } n \text{ is odd,} \\ n/2, & \text{if } n \text{ is even.} \end{cases}$$

The Collatz problem has been studied from several viewpoints, including analytic estimates, probabilistic arguments, computational verification, and structural reformulations. A comprehensive overview of the problem and its major developments is given by Lagarias [1], who discusses analytic estimates, probabilistic models, computational verification, and structural interpretations of Collatz trajectories. Krasikov and Lagarias [2] developed analytic bounds using difference inequalities, showing how probabilistic and inequality-based methods can yield nontrivial information about the distribution of Collatz trajectories. Tao [3] proved that almost all orbits of the Collatz map attain almost bounded values, providing a major probabilistic advance in understanding typical orbit behavior. Another important line of research focuses on the structure of trajectories through cycle lengths, directed graphs, and finite state representations. Eliahou [4] obtained lower bounds on nontrivial cycle lengths, contributing to the study of possible periodic behavior in Collatz dynamics. From a graph-theoretic viewpoint, Andaloro and, later, van Laarhoven and de Weger investigated graph-based descriptions of Collatz trajectories, including directed graph models and connections with De Bruijn graphs, making it possible to study

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their iterative behavior through combinatorial and dynamical patterns [5,6]. Conway's (see [7]) foundational work highlighted the complex and potentially unpredictable nature of generalized Collatz iterations, showing how simple recursive rules can generate highly nontrivial computational behavior. A clustering perspective on the Collatz conjecture has also been proposed to study [8] structural regularities among Collatz trajectories and to highlight pattern-based relationships in their iterative behavior. Motivated by these structural approaches, we study the level sets l_x through the parity patterns of the corresponding trajectories.

Starting from a positive integer n, the associated Collatz sequence is

$$n,\ f(n),\ f^{(2)}(n),\ f^{(3)}(n), \dots,$$

where $f^{(k)}(n)$ denotes the k-th iterate of f. The Collatz conjecture asserts that for every $n \in \mathbb{N}$, there exists a finite integer k such that

$$f^{(k)}(n) = 1.$$

The smallest such k is called the Collatz length of n, and we denote it by

$$L(n) = \min\{\, k \in \mathbb{N} \mid f^{(k)}(n) = 1 \,\}.$$

For each integer $x \geq 1$, we define the level set

$$l_x = \{\, n \in \mathbb{N} \mid L(n) = x \,\},$$

which consists of all positive integers whose trajectories reach 1 after exactly x iterations.

Level sets of Collatz lengths have been investigated from various perspectives, including probabilistic models, analytic bounds, and large-scale computational verification [9–12]. In this paper, we study the structure of the sets l_x through the parity patterns of the corresponding Collatz trajectories.

For each integer n with Collatz length $L(n) = x$, we associate a parity vector

$$p(n) = (p_0, p_1, \dots, p_{x-1}) \in \{0,1\}^x,$$

where

$$p_t = \begin{cases} 1, & \text{if the } t\text{-th iterate is odd,} \\ 0, & \text{if the } t\text{-th iterate is even.} \end{cases}$$

Thus, each element of l_x is encoded by a binary sequence of length x that records the parity structure of its trajectory.

A fundamental quantity derived from this representation is the number of odd steps,

$$k(n) = \sum_{t=0}^{x-1} p_t,$$

which counts how many times the odd transformation $3n + 1$ occurs in the trajectory of n. For a fixed length x, this leads naturally to the subsets

$$\emptyset \neq S_{x,k} = \{\, n \in l_x : k(n) = k \,\},$$

which partition the level set l_x according to odd-step count. We denote the mean of each such subset by

$$\mu_{x,k} = \frac{1}{|S_{x,k}|} \sum_{n \in S_{x,k}} n.$$

The main idea of this paper is that the parity representation provides a natural framework for understanding the numerical distribution of integers inside each level set l_x. Heuristically, if a trajectory of length x contains k odd steps, then its starting value is expected to scale approximately like

$$n \approx C(p)\,\frac{2^x}{6^k},$$

where C depends on the specific parity pattern of the trajectory. This suggests that the means $\mu_{x,k}$ should exhibit an approximate geometric behavior as functions of k.

Motivated by this observation, we develop a statistical analysis of the parity classes $S_{x,k}$. We prove a basic mean bound under a natural scaling hypothesis, derive bounds for the successive ratios $\mu_{x,k+1}/\mu_{x,k}$, and present numerical results showing that the logarithms of these means are very well approximated by linear functions of k. These results indicate that the odd-step count is a useful structural parameter for describing the distribution of integers within the level sets l_x.

For $n \in l_x$, let

$$0 \le t_1 < t_2 < \cdots < t_k \le x-1$$

denote the positions of the odd steps in the parity code of n. It is convenient to introduce the trajectory-dependent quantity

$$C(p) = \frac{n\,6^k}{2^x},$$

which depends only on the parity pattern p of the trajectory. The following lemma gives an exact expression for n and hence for $C(p)$.

Lemma 1. *Let $n \in l_x$, and let $t_1 < t_2 < \cdots < t_k$ be the positions of the odd steps in its parity code. Then,*

$$n = \frac{2^{x-k}}{3^k} - \sum_{r=1}^{k} \frac{2^{\,t_r+1-r}}{3^r}.$$

Equivalently,

$$n = C(p)\,\frac{2^x}{6^k},$$

where

$$C(p) = 1 - \sum_{r=1}^{k} 2^{\,t_r+1-r+k-x}\,3^{\,k-r}.$$

Proof. Let

$$a_0 = n, \qquad a_{j+1} = f(a_j), \qquad a_x = 1$$

be the Collatz trajectory of n.

At each step, the update has the form

$$a_{j+1} = \begin{cases} 3a_j + 1, & \text{if } p_j = 1, \\ a_j/2, & \text{if } p_j = 0. \end{cases}$$

Hence, every odd step contributes a multiplicative factor 3 and an additive term 1, while every even step contributes a factor $1/2$.

If the parity code contains exactly k odd steps, then the total multiplicative contribution from $a_0 = n$ to a_x is

$$\frac{3^k}{2^{x-k}}\, n.$$

Now consider the additive contribution coming from the r-th odd step, which occurs at position t_r. After this step, there remain exactly $k - r$ odd steps and

$$x - 1 - t_r - (k - r)$$

even steps. Therefore, the additive term 1 generated at position t_r contributes

$$\frac{3^{k-r}}{2^{\,x-1-t_r-(k-r)}}$$

to the final value a_x.

Since $a_x = 1$, summing the multiplicative and additive contributions gives

$$1 = \frac{3^k}{2^{x-k}}\, n + \sum_{r=1}^{k} \frac{3^{k-r}}{2^{\,x-1-t_r-(k-r)}}.$$

Multiplying both sides by $\frac{2^{x-k}}{3^k}$, we obtain

$$n = \frac{2^{x-k}}{3^k} - \sum_{r=1}^{k} \frac{2^{x-k}}{3^k} \cdot \frac{3^{k-r}}{2^{\,x-1-t_r-(k-r)}}.$$

Simplifying the exponents yields

$$n = \frac{2^{x-k}}{3^k} - \sum_{r=1}^{k} \frac{2^{\,t_r+1-r}}{3^r}.$$

Finally, since

$$\frac{2^{x-k}}{3^k} = \frac{2^x}{6^k},$$

we may factor out $\frac{2^x}{6^k}$ and write

$$n = C(p)\, \frac{2^x}{6^k},$$

where

$$C(p) = \frac{n\, 6^k}{2^x}.$$

Substituting the explicit formula for n gives

$$C(p) = 1 - \sum_{r=1}^{k} 2^{\,t_r+1-r+k-x}\, 3^{\,k-r}.$$

□

The rest of the paper is organized as follows. In Section 2, we introduce the parity classes $S_{x,k}$ and establish the main theoretical bounds for their means. In Section 3, we present examples and numerical results illustrating the stability of these mean ratios and the corresponding regression behavior across a range of Collatz lengths.

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2. Parity Representation and Statistical Clustering

In addition to studying the numerical distribution of integers within each level set,

$$l_x = \{n \in \mathbb{N} \mid L(n) = x\},$$

we examine the internal structure of Collatz trajectories using a binary feature representation.

Let

$$a_0 = n, \qquad a_{t+1} = f(a_t)$$

be the Collatz trajectory of an integer n with length $L(n) = x$. We define the *parity vector*

$$p(n) = (p_0, p_1, \ldots, p_{x-1}) \in \{0,1\}^x$$

by

$$p_t = \begin{cases} 1, & a_t \text{ is odd,} \\ 0, & a_t \text{ is even.} \end{cases}$$

Thus, each integer $n \in l_x$ can be associated with a binary sequence describing the parity pattern of its Collatz trajectory. The length of this sequence equals the Collatz length x.

Example 1. *For Collatz length* $x = 12$*, the following integers and their parity codes arise:*

n	$L(n)$	Parity Code
17	12	100100010000
96	12	000001010000
104	12	000100010000
106	12	010000010000
113	12	100100000000
640	12	000000010000
672	12	000001000000
680	12	000100000000
682	12	010000000000
4096	12	000000000000

A natural feature derived from the parity vector is the number of odd steps,

$$k(n) = \sum_{t=0}^{x-1} p_t,$$

which counts the total number of occurrences of the odd transformation $3n + 1$ in the trajectory of n.

In the above example, the elements of l_{12} are grouped according to the value of $k(n)$ as follows:

$$k = 3 : \{17\},$$
$$k = 2 : \{96, 104, 106, 113\},$$
$$k = 1 : \{640, 672, 680, 682\},$$
$$k = 0 : \{4096\}.$$

Thus, the number of odd steps provides a natural way to organize the integers in l_x.

This can be understood heuristically from the multiplicative behavior of the Collatz map. If a trajectory of length x contains k odd steps and $x - k$ even steps, then each odd step contributes approximately a factor of 3, while each even step contributes a factor of

1/2. Ignoring the additive $+1$ terms in the odd steps, the overall effect on the initial value is therefore approximately

$$n \longmapsto \frac{3^k}{2^{x-k}}\, n = \frac{6^k}{2^x}\, n.$$

Since the trajectory eventually reaches 1, this suggests the approximation

$$1 \approx \frac{6^k}{2^x}\, n,$$

and hence,

$$n \approx C\, \frac{2^x}{6^k},$$

where C depends on the precise parity pattern of the trajectory. The constant C accounts for the cumulative effect of the additive $+1$ terms in the odd steps as well as the ordering of odd and even operations.

Taking logarithms gives

$$\log n \approx x \log 2 - k \log 6 + \log C,$$

which shows that, for fixed length x, the logarithm of n should vary approximately linearly with the number of odd steps k.

This provides a statistical explanation for the geometric behavior of the means $\mu_{x,k}$. Since integers with similar values of k tend to lie in comparable numerical ranges, the means of the classes $S_{x,k}$ are expected to exhibit approximately constant ratios. Thus, the observed geometric behavior may be viewed as a consequence of the underlying parity structure of Collatz trajectories.

Fix $x \geq 1$ and consider the level set

$$l_x = \{n \in \mathbb{N} \mid L(n) = x\}.$$

For each $n \in l_x$, let $k(n)$ denote the number of odd steps in its Collatz trajectory.

For each admissible value k, define the subset

$$S_{x,k} = \{\, n \in l_x : k(n) = k \,\}.$$

Thus, the level set decomposes as

$$l_x = \bigcup_k S_{x,k}.$$

For each k, we define the mean value of the set $S_{x,k}$ by

$$\mu_{x,k} = \frac{1}{|S_{x,k}|} \sum_{n \in S_{x,k}} n.$$

Lemma 2. *Assume there exist constants $A_x, B_x > 0$ such that for every admissible k and every $n \in S_{x,k}$,*

$$A_x \frac{2^x}{6^k} \leq n \leq B_x \frac{2^x}{6^k}.$$

Then, the mean value $\mu_{x,k}$ satisfies

$$A_x \frac{2^x}{6^k} \leq \mu_{x,k} \leq B_x \frac{2^x}{6^k}.$$

Proof. Fix k. By assumption, for every $n \in S_{x,k}$,

$$A_x \frac{2^x}{6^k} \leq n \leq B_x \frac{2^x}{6^k}.$$

Summing over all $n \in S_{x,k}$ yields

$$|S_{x,k}|\, A_x \frac{2^x}{6^k} \leq \sum_{n \in S_{x,k}} n \leq |S_{x,k}|\, B_x \frac{2^x}{6^k}.$$

Dividing by $|S_{x,k}|$, we obtain

$$A_x \frac{2^x}{6^k} \leq \frac{1}{|S_{x,k}|} \sum_{n \in S_{x,k}} n \leq B_x \frac{2^x}{6^k}.$$

□

Remark 1. *Recall that*

$$n = C(p)\frac{2^x}{6^k}, \qquad C(p) = \frac{n\, 6^k}{2^x},$$

where $C(p)$ depends on the parity pattern p of the trajectory. If one wishes to determine the sharpest constants A_x and B_x satisfying the hypothesis of Lemma 2, then they are given by

$$A_x = \min_k \min_{n \in S_{x,k}} C(p) = \min_k \min_{n \in S_{x,k}} \frac{n\, 6^k}{2^x},$$

and

$$B_x = \max_k \max_{n \in S_{x,k}} C(p) = \max_k \max_{n \in S_{x,k}} \frac{n\, 6^k}{2^x}.$$

Indeed, the inequalities

$$A_x \frac{2^x}{6^k} \leq n \leq B_x \frac{2^x}{6^k} \qquad \text{for all } n \in S_{x,k}$$

are equivalent to

$$A_x \leq C(p) \leq B_x \qquad \text{for all } n \in S_{x,k}.$$

Thus, the optimal lower and upper bounds are obtained by taking the minimum and maximum of $C(p)$ over all admissible pairs (k, n).

Example 2. *Consider the level set*

$$l_{12} = \{17, 96, 104, 106, 113, 640, 672, 680, 682, 4096\}.$$

Grouping by odd-step count gives

$$S_{12,3} = \{17\}, \quad S_{12,2} = \{96, 104, 106, 113\}, \quad S_{12,1} = \{640, 672, 680, 682\}, \quad S_{12,0} = \{4096\}.$$

Their means are

$$\mu_{12,3} = 17, \qquad \mu_{12,2} = 104.75, \qquad \mu_{12,1} = 668.5, \qquad \mu_{12,0} = 4096.$$

If we use the scaling law $n \approx C\, 2^x / 6^k$, then the optimal constants are

$$A_{12} = \min_k \min_{n \in S_{12,k}} \frac{n\, 6^k}{2^{12}} = \frac{27}{32}, \qquad B_{12} = \max_k \max_{n \in S_{12,k}} \frac{n\, 6^k}{2^{12}} = 1.$$

Hence, for every admissible k and every $n \in S_{12,k}$,

$$A_{12}\frac{2^{12}}{6^k} \leq n \leq B_{12}\frac{2^{12}}{6^k}.$$

Therefore,

$$A_{12}\frac{2^{12}}{6^k} \leq \mu_{12,k} \leq B_{12}\frac{2^{12}}{6^k}.$$

For instance, when $k = 2$,

$$\frac{27}{32} \cdot \frac{2^{12}}{6^2} = 96 \leq \mu_{12,2} = 104.75 \leq \frac{2^{12}}{6^2} \approx 113.78.$$

Corollary 1. *Suppose the hypothesis of Lemma 2 holds. Then, for every k,*

$$\frac{A_x}{6B_x} \leq \frac{\mu_{x,k+1}}{\mu_{x,k}} \leq \frac{B_x}{6A_x}.$$

Proof. By Lemma 2,

$$A_x\frac{2^x}{6^k} \leq \mu_{x,k} \leq B_x\frac{2^x}{6^k}$$

and

$$A_x\frac{2^x}{6^{k+1}} \leq \mu_{x,k+1} \leq B_x\frac{2^x}{6^{k+1}}.$$

Dividing the second inequality by the first gives

$$\frac{A_x 2^x/6^{k+1}}{B_x 2^x/6^k} \leq \frac{\mu_{x,k+1}}{\mu_{x,k}} \leq \frac{B_x 2^x/6^{k+1}}{A_x 2^x/6^k},$$

which simplifies to

$$\frac{A_x}{6B_x} \leq \frac{\mu_{x,k+1}}{\mu_{x,k}} \leq \frac{B_x}{6A_x}.$$

□

Theorem 1. *Suppose the hypothesis of Lemma 2 holds. Then, for every k,*

$$\left|\frac{\mu_{x,k+2}}{\mu_{x,k+1}} - \frac{\mu_{x,k+1}}{\mu_{x,k}}\right| \leq \frac{B_x}{6A_x} - \frac{A_x}{6B_x}.$$

Proof. By Corollary 1, both ratios

$$\frac{\mu_{x,k+2}}{\mu_{x,k+1}} \quad \text{and} \quad \frac{\mu_{x,k+1}}{\mu_{x,k}}$$

belong to the interval

$$\left[\frac{A_x}{6B_x}, \frac{B_x}{6A_x}\right].$$

Hence, the absolute difference between them is bounded by the length of this interval; then, we have

$$\left|\frac{\mu_{x,k+2}}{\mu_{x,k+1}} - \frac{\mu_{x,k+1}}{\mu_{x,k}}\right| \leq \frac{B_x}{6A_x} - \frac{A_x}{6B_x}.$$

□

Statistical Model for Parity-Based Clusters

The parity representation introduced in the previous section suggests that the integers in the set

$$l_x = \{n \in \mathbb{N} \mid L(n) = x\}$$

can be naturally grouped according to the number of odd steps in their Collatz trajectories.

For a fixed Collatz length x, define

$$k(n) = \sum_{t=0}^{x-1} p_t,$$

where p_t denotes the parity indicator defined earlier. Thus, $k(n)$ counts the total number of odd steps in the trajectory of n.

The empirical analysis indicates that these means follow an approximate exponential relationship with respect to k. Specifically, the data suggest a model of the form

$$\mu_k = Ae^{bk},$$

where $A > 0$ and $b < 0$ are constants depending on the Collatz length x.

Taking natural logarithms yields the linear relation

$$\log \mu_k = \log A + bk.$$

Thus, the parameters A and b can be estimated by fitting a linear regression model to the points

$$(k, \log \mu_k).$$

3. Numerical Results

To investigate the statistical structure of the level sets l_x, we computed all integers

$$n \leq 100{,}000$$

and determined their Collatz lengths. For each length $x = 10, 11, \ldots, 50$, the set

$$l_x = \{n \in \mathbb{N} \mid L(n) = x\}$$

was constructed. Integers in l_x were then grouped according to the number of odd steps

$$k(n) = \sum_{t=0}^{x-1} p_t$$

in their parity representation. The regression analysis is used to test whether the relationship between the cluster means μ_k and the number of odd steps k follows an exponential pattern. Accordingly, for each cluster indexed by k, the mean value μ_k was computed, and a linear regression of

$$\log \mu_k$$

against k was performed. The resulting regression model has the form

$$\log \mu_k = a + bk.$$

To summarize these numerical findings, Table 1 and Figures 1 and 2 present the behavior of the regression parameters across Collatz lengths.

Table 1 shows that, across the tested lengths, the parity-based clustering is consistently pure (purity $= 1.0$), meaning that each cluster is perfectly separated by the odd-step count in the computed data. Across all tested values of L, the linear fits were extremely strong, with coefficients of determination R^2 values that are extremely close to 1, indicating that the logarithm of the cluster means is very well approximated by a linear function of the cluster index. Moreover, the fitted slopes b are highly stable, $(-1.80 \leq b \leq -1.86)$. This supports the claim that the cluster means of the parity classes follow an approximate exponential trend as the odd-step count increases.

Figure 1 shows the estimated slope b as a function of the Collatz length L. The graph demonstrates that the slope remains highly stable across the entire range $10 \leq L \leq 50$.

Figure 2 illustrates the growth of the level-set size $|l_L|$ as L increases, while exhibiting fluctuations likely tied to the irregular structure of Collatz trajectories. This provides context for the regression analysis by showing how many integers contribute to each computation.

Table 1. Regression summary for Collatz level sets of length L showing the level-set of size $|l_L|$, the number of parity clusters, the purity of the clustering, the fitted exponential slope b, and the coefficient of determination R^2.

L	$\|l_L\|$	Clusters	Purity	b	R^2
10	6	3	1.0	−1.8563	0.9999
11	8	3	1.0	−1.8464	0.9999
12	10	4	1.0	−1.8307	1.0000
13	14	4	1.0	−1.8251	1.0000
14	18	5	1.0	−1.8252	1.0000
15	24	5	1.0	−1.8198	1.0000
16	29	6	1.0	−1.8274	1.0000
17	35	5	1.0	−1.8259	1.0000
18	43	5	1.0	−1.8248	1.0000
19	57	6	1.0	−1.8314	0.9999
20	63	5	1.0	−1.8337	1.0000
21	81	5	1.0	−1.8317	0.9999
22	80	5	1.0	−1.7969	0.9995
23	105	5	1.0	−1.8315	1.0000
24	138	5	1.0	−1.8302	1.0000
25	124	4	1.0	−1.8368	1.0000
26	171	5	1.0	−1.8314	1.0000
27	186	5	1.0	−1.8217	0.9999
28	195	4	1.0	−1.8359	1.0000
29	268	5	1.0	−1.8338	1.0000
30	201	4	1.0	−1.8336	1.0000
31	289	4	1.0	−1.8330	1.0000
32	416	5	1.0	−1.8275	1.0000
33	290	4	1.0	−1.8305	1.0000
34	428	5	1.0	−1.8249	1.0000
35	261	4	1.0	−1.8244	1.0000
36	408	4	1.0	−1.8269	1.0000
37	618	4	1.0	−1.8285	1.0000
38	371	4	1.0	−1.8214	1.0000
39	571	4	1.0	−1.8252	1.0000
40	629	4	1.0	−1.8090	0.9999
41	483	4	1.0	−1.8250	1.0000
42	782	4	1.0	−1.8187	1.0000
43	400	3	1.0	−1.8227	1.0000
44	652	4	1.0	−1.8153	1.0000
45	1050	4	1.0	−1.8189	1.0000
46	510	4	1.0	−1.8124	1.0000
47	863	4	1.0	−1.8136	1.0000
48	384	3	1.0	−1.8146	1.0000
49	672	4	1.0	−1.8065	1.0000
50	1148	4	1.0	−1.8056	1.0000

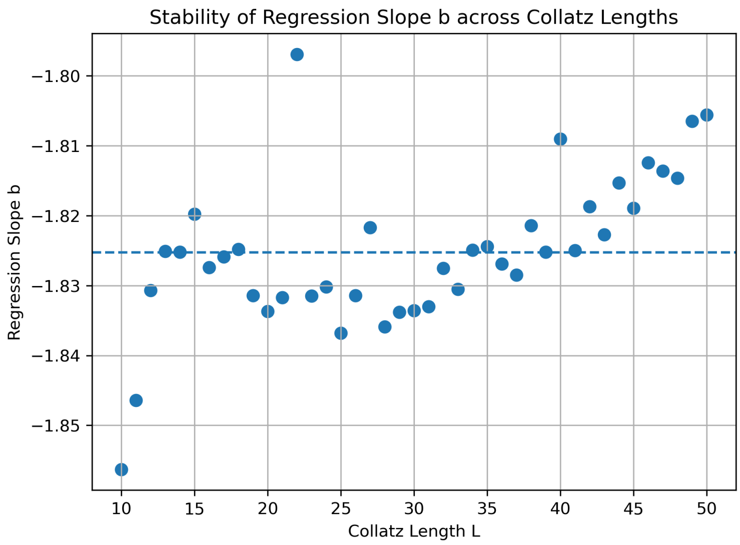

Figure 1. Estimated regression slope b as a function of Collatz length L. The slope remains highly stable across the tested range $10 \leq L \leq 50$, indicating that the exponential decay rate of cluster means is essentially constant.

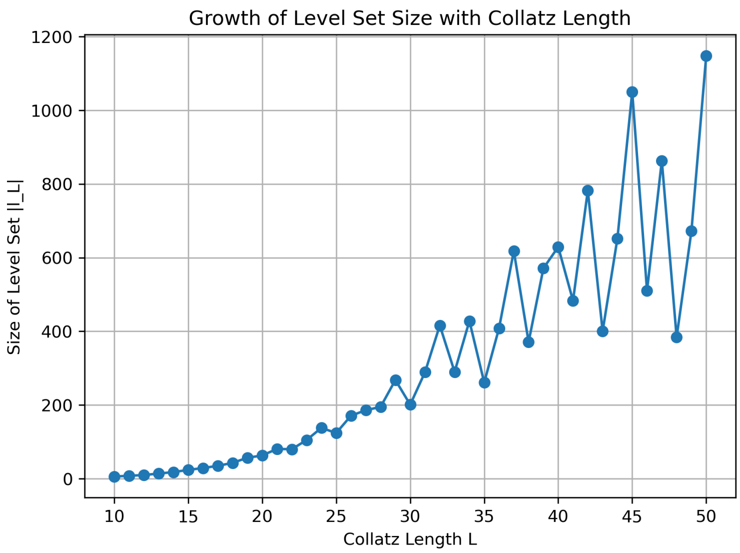

Figure 2. Growth of the level-set size $|l_L|$ with respect to the Collatz length L.

4. Conclusions

In this paper, we studied the level sets

$$l_x = \{n \in \mathbb{N} \mid L(n) = x\}$$

of the Collatz map through the parity structure of their trajectories. For each fixed length x, we partitioned l_x into the subsets

$$S_{x,k} = \{n \in l_x : k(n) = k\},$$

where $k(n)$ denotes the number of odd steps in the trajectory of n, and we analyzed the corresponding means

$$\mu_{x,k} = \frac{1}{|S_{x,k}|} \sum_{n \in S_{x,k}} n.$$

The parity-based viewpoint developed here suggests several possible directions for further study. One is to investigate whether sharper theoretical bounds for A_x and B_x

can be obtained directly from the parity patterns of Collatz trajectories. Another is to examine whether similar statistical laws persist for larger computational ranges or for other quantities associated with the trajectories. More broadly, the results indicate that the level sets l_x possess additional internal structure that may be useful in understanding the dynamics of the Collatz map.

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